Sesha Sai Aditya Penumaka

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Professional Summary

AI/ML Engineer and Certified AWS Cloud Native Developer with experience at Accenture and Cognizant. Specializing in Generative AI, MLOps, and Hybrid Neural Networks. Proven track record in developing scalable AI systems and conducting research in Entity Resolution, aiming to advance the field of Large Language Models and production-grade AI through graduate studies.

Education

CMR College of Engineering and Technology
Bachelor of Technology (AI & DS) |2021-2025
Narayana Jr. College (MPC) |2019 – 2021

Experience

Cognizant Technology Solutions Corporation — June 2025 – Feb 2026

- **Developed end-to-end Machine Learning models** (Supervised/Unsupervised) to solve high-impact business problems, focusing on predictive analytics and data-driven decision-making.
- **Engineered robust data preprocessing** and model optimization strategies, utilizing SMOTE for imbalanced datasets and **XGBoost** to boost model sensitivity and precision.
- **Collaborated on hybrid AI frameworks** involving **CNNs** and **LSTMs**, contributing to the development of automation tools that streamlined internal data categorization and reporting.

Accenture Solutions Private Limited — Feb 2026 – Present

- **Architected and deployed** scalable deep learning pipelines using **TensorFlow** and **Keras, integrating Hybrid Neural Networks to enhance predictive accuracy in complex datasets.**
- **Optimized large-scale NLP models** by implementing advanced feature engineering and transformer-based architectures, reducing computational latency while maintaining high precision for real-time sentiment analysis.
- **Led the integration of Computer Vision** solutions using **OpenCV**, automating image classification tasks that improved data processing efficiency across cross-functional workflows.

Projects

Cancer Cell Detection using Hybrid Neural Networks (CCDC-HNN) — Dec 2024 – Jan 2025

- **Developed a hybrid deep learning architecture** combining **CNNs** for spatial feature extraction and **LSTMs** to analyze sequential cellular growth patterns.
- **Optimized model performance** using **OpenCV** for image augmentation and preprocessing, achieving a significant reduction in false negatives for early-stage malignancy detection.
- **Implemented custom loss functions** and fine-tuned hyperparameters in **TensorFlow**, ensuring the model's robustness against high-variance clinical datasets.

Credit Card Fraud Detection using XGBoost — Nov 2024 – Jan 2025

- **Architected a real-time fraud identification system** using **Python** and **XGBoost,** focusing on minimizing financial risk through high-precision classification.
- **Mitigated extreme class imbalance** by implementing **SMOTE** (Synthetic Minority Over-sampling Technique) and advanced feature engineering, boosting the F1-score for minority fraud classes.

Technical Skills

- **Programming & Tools:** Python, SQL, Git & GitHub, Jupyter Notebook, Google Colab, Docker, MLFlow
- **Skills:** Supervised & Unsupervised Learning, Generative AI, RAG, CNN, RNN, Hybrid Neural Networks, NLP, Agentic AI, Fine-tuning
- **Frameworks & Libraries:** TensorFlow, Keras, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Certifications

Machine learning Algorithms Certification.
MySql Certification